

Data Pipeline Specification — Game Title

Fictional demo document for Studio Agent Nexus. Synthetic content, safe for public demonstration. Specifies how gameplay and platform events are ingested, processed, and stored for analytics at Game Studio.

1. Purpose

This spec defines the event pipeline that carries telemetry from game servers and platform services into the analytics warehouse. The pipeline is asynchronous and never sits on the gameplay hot path.

2. Event flow

Producers (game servers and services) publish events to a message bus. Stream processors validate, enrich, and route events into warehouse tables. Consumers include analytics, live-ops tooling, and anti-cheat.

- Transport: a partitioned message bus with per-domain topics.
- Topics: match_events, economy_events, auth_events, session_events, ad_events.
- Delivery: at-least-once; consumers are idempotent on event_id.

3. Event schema

Every event shares an envelope: event_id, event_type, player_id, session_id, server_id, timestamp (UTC), schema_version, and a typed payload. Schemas are versioned; breaking changes require a new schema_version and a migration note.

4. Sampling and volume

High-frequency gameplay signals are sampled before publication to keep volume bounded; low-frequency lifecycle events (match start/end, purchases, logins) are always published. Sampling rates are managed through live config.

5. Warehouse and retention

Validated events land in the analytics warehouse, partitioned by event_type and date. Retention windows are defined per table. Daily metrics are bucketed by UTC calendar day; for example, D1 retention is computed against the next calendar day after install, not a rolling 24-hour window.

Open gap: the retention and purge window for player-identifying (PII) event fields is not specified. Until a compliance decision sets an explicit purge window, PII-bearing events accumulate without a defined deletion policy.

6. Quality and monitoring

- Schema validation rejects malformed events to a dead-letter topic.
- Ingestion lag and dead-letter volume are dashboarded and alerted.

- Lossless ingestion at expected volume is a beta go-live criterion.

7. Access

Analytics consumers read curated warehouse tables, not raw topics. Raw topic access is restricted to the Data team. Ownership is recorded in the Team Directory.